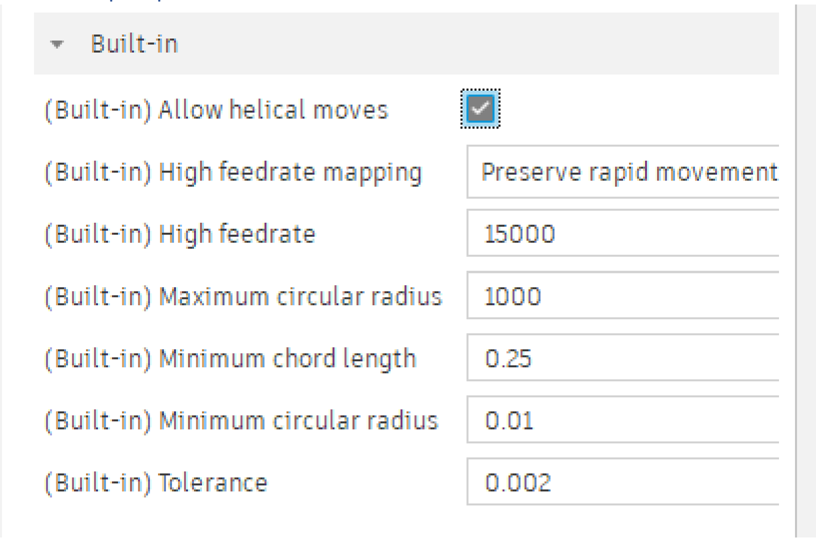
Fusion Post Processor Information

Description of the post properties

Built-in properties

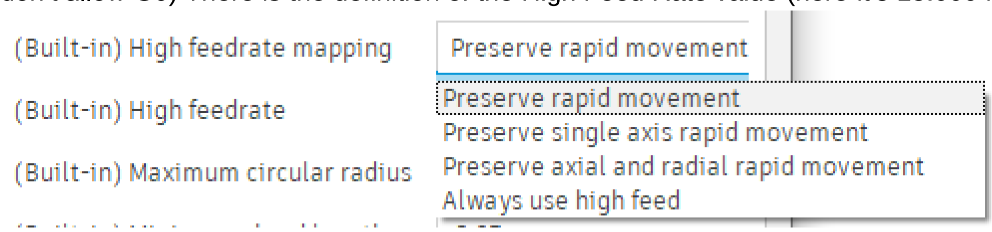


Arc and helix interpolation

There are six parameters managing the helix and arcs generation. This first one is to *allow the helical moves* output.
We have the filtering values for the *maximum* and *minimum* arc radii, handled by the post. The other dimensions will be linearized using the *Tolerance*. Another filter is the *minimum chord length*.
The *tolerance* is not the toolpath tolerance, but the tolerance used to linearize arcs, helix, or curve when applicable.

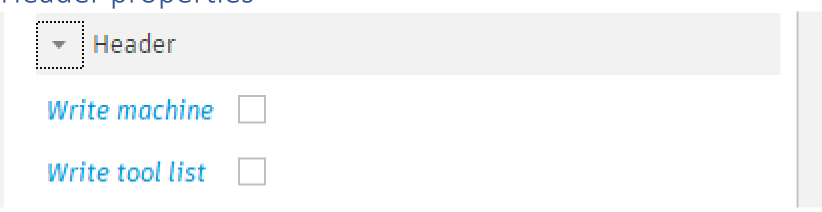
High Feed Rate

There are 2 parameters managing the high feed rate, but it also depends on your license. (Personal account don't allow G0) There is the definition of the High Feed Rate value (here it's 15.000 mm/min)



The other parameter is used for mapping the rapid moves, preserving them, converting them to high feed rate…

Header properties



These switches handle the output of some comments from different sources.

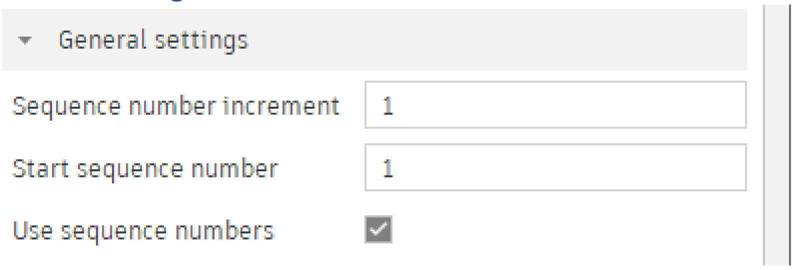
The *Write machine* will get the brand, model name and description of the machine from the machine configuration file selected in the setup. Can be useful if you use a generic post processor for several machines. The output looks like :

```
(4444)
(MACHINE)
(  VENDOR AUTODESK)
(  MODEL GENERIC 5-AXIS CA TABLE-HEAD)
(  DESCRIPTION THIS MACHINE HAS XC AXIS ON THE TABLE AND YZA AXIS ON THE HEAD)
N1 G90
N2 G17 Q1
N3 G71
N4 M11
```

The *Write tool list* option will create a list of tools with the length, diameter and max depth reached by the tool, in the header at the beginning of the g-code.

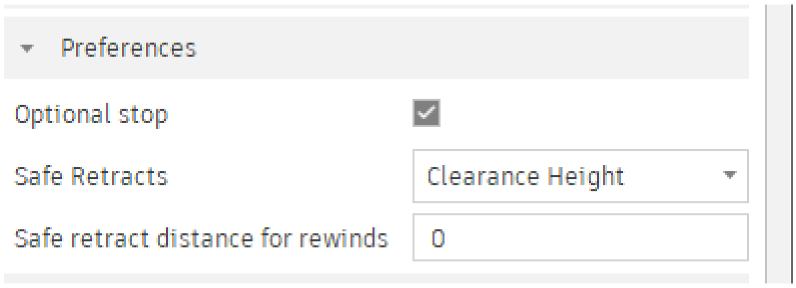
```
(4444)
(T01  D=50. CR=0. - ZMIN=-1. - FACE MILL)
N1 G90
N2 G17 Q1
N3 G71
```

General Settings



These parameters are managing the line numbers. We can define the *Start sequence number*, the *Sequence number increment*. And the *Use sequence numbers* allow to switch line numbering on and off.

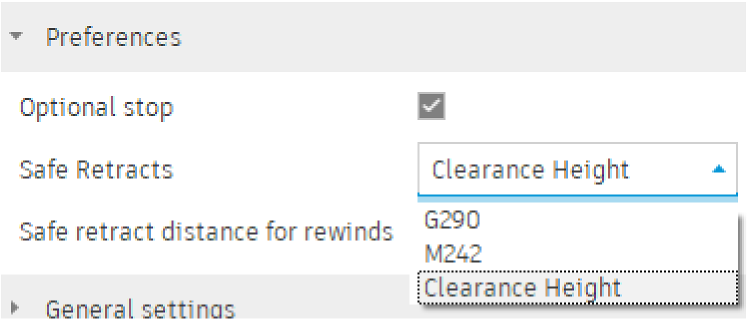
Preferences



The *Optional stop* switch allow to output an M1 command between toolpaths as in :

```
N17 X-80.
N18 G02 Y0.95 I-80. J-11.712
N19 G01 X80.
N20 G00 Z15.
(2D-CONTOUR)
N21 M09
N22 M01
N23 M06 T.02
N24 S6000 M03
N25 G55 O1
```

The *Safe retract distance* for rewinds will define the length of the retract, along the tool axis, when we need to perform a rewind. Rewind occurs when a rotary axis reaches one of his limits and we need to switch to an alternate positioning.



The *Safe retract* allow us to select the retraction mode between operation.
There is three choices *Clearance height* only useable for 3 axis toolpaths / machine. Then the M242 which is retracting the machine in Z (possibility only on 3 axis machine), And the *G290*, more secure for a multi axis machine, if available.(CNC software over 5.7 support this, v3.9 does not)

Multi-axis

These two switches will tweak the behavior when outputting 5 axis positioned toolpaths.
Preposition rotaries will force a move of the rotary axis before eventually outputting a tilted work plane command. It sometime prevents the machine from selecting an alternate positioning.
Use G21 allow to output a G21 command with the Euler angle corresponding to the tilted work plane orientation. A tilted work plane can allow cutter compensation and canned cycles with any machine kinematic.

Reach to us via [HSM Post Processor Forum - Autodesk Community] if you have any issues